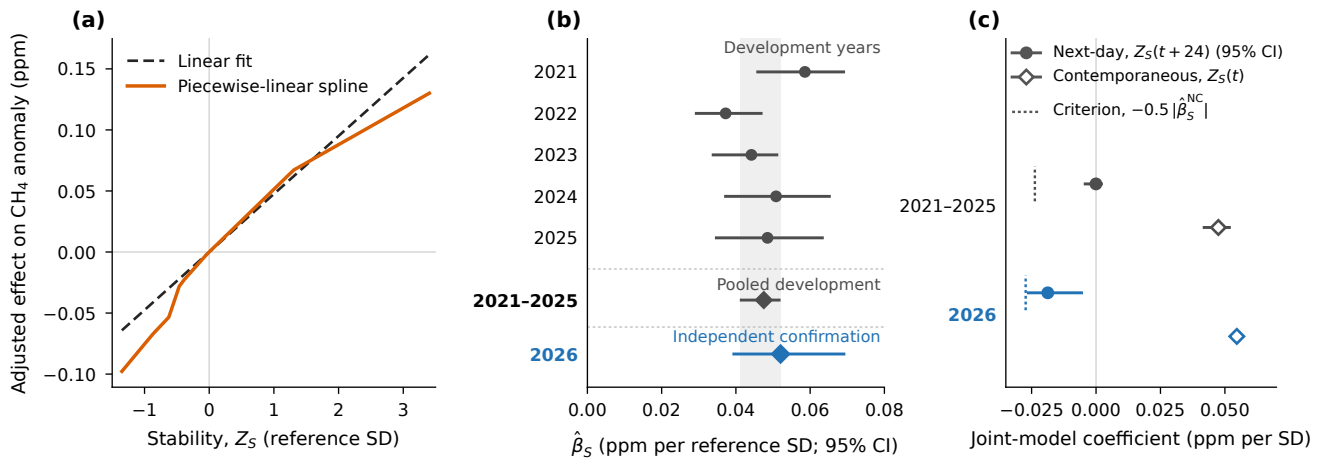
(a)
Linear fit
Piecewise-linear spline
Adjusted effect on CH4 anomaly (ppm)
0.15
0.10
0.05
0.00
−0.05
−0.10
−1
0
1
2
3
Stability, Z_S (reference SD)
(b)
Development years
2021
2022
2023
2024
2025
Pooled development
2021–2025
Independent confirmation
2026
0.00
0.02
0.04
0.06
0.08
$\hat{\beta}_S$ (ppm per reference SD; 95% CI)
(c)
Next-day, $Z_S(t+24)$ (95% CI)
Contemporaneous, $Z_S(t)$
Criterion, $-0.5\,|\hat{\beta}_S^{NC}|$
2021–2025
2026
−0.025
0.000
0.025
0.050
Joint-model coefficient (ppm per SD)